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 Applicability UMHS Clinical
 References Guideline

Pharmacy Services Adult Continuous Infusions in the Intensive Care Unit, Adult Emergency Services, UH & CVC PACUs and Progressive Care Units (8D, 4AS & CVC-5) Guideline

Guideline overview, Purpose, and Scope

To provide guidance on the standard concentration, starting dose, titration amount/frequency, and maximum doses of continuous infusion medications. These standards apply to most situations, but may be departed from during exceptional circumstances on a case-by-case basis with clinical judgment and collaboration among the multi-disciplinary team.

Definitions

None

Areas of Responsibility

Nurses, Prescribers, and Pharmacists in Intensive Care Units, Adult Emergency Services, UH & CVC PACUs, and Progressive Care Units

Guideline Steps

SEDATIVE AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
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Dexmedetomidine (Precedex)	200 mcg/50 mL (4 mcg/mL)	Start: 0.2-0.5 mcg/kg/hour Titrate: 0.1-0.2 mcg/kg/hour every 15 minutes	CICU: 0.7 mcg/kg/hour 1.5 mcg/kg/hour	Central or peripheral administration Restricted to ICU and AES
Haloperidol (Haldol)	1 mg/mL	Start: 4-10 mg/hour	10 mg/hour	Central or peripheral administration Restricted to ICU and AES Obtain a baseline and daily EKG and consider additional EKGs as required
Ketamine	10 mg/mL	Start: 0.2-0.6 mg/kg/hour Titrate: 0.1 mg/kg/hour every 15 minutes	Sedation: 1.2 mg/kg/hour Status Asthmaticus: 2.5 mg/kg/hr	Central or peripheral administration
LORazepam (Ativan)	60 mg/30 mL (2 mg/mL)	Start: 1-10 mg/hour (initial dose should be based upon dose requirements over previous 4 hours) Titrate: 20-50% every 30 minutes	20 mg/hour	Central or peripheral administration Restricted to ICU and AES
Midazolam (Versed)	150 mg/30 mL (5 mg/mL)	Start: 2-10 mg/hour Titrate: 20-50% every 10 minutes	30 mg/hour	Central or peripheral administration Restricted to ICU and AES
Propofol (Diprivan)	1,000 mg/100 mL 500 mg/50 mL (10 mg/mL)	Start: 5-20 mcg/kg/min Titrate: 5-20 mcg/kg/min. every 2 minutes	80 mcg/kg/min	Central or peripheral administration Restricted to ICU and AES Monitor serum triglyceride concentration within 24-48 hours of initiation and repeat every 2-5 days
ANALGESIC AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES

FentaNYL	1,500 mcg/30 mL (50 mcg/mL)	Start: 12.5-50 mcg/hour Titrate: 25 mcg/hour every 5 minutes	200 mcg/hour	Central or peripheral administration
HYDROmorphine (Dilaudid)	75 mg/30 mL (2.5 mg/mL)	Start: 0.25-0.5 mg/hour Titrate: 0.25-0.5 mg/hour every 30 minutes	3 mg/hour	Central or peripheral administration
Morphine Sulfate	150 mg/30 mL (5 mg/mL)	Start: 2-4 mg/hour (initial dose should be based upon dose requirements over previous 4 hours) Titrate: 1-2 mg/hour every 30 minutes	20 mg/hour	Central or peripheral administration <i>These parameters may not apply to the management of symptoms at the end of life</i>
NEURO-MUSCULAR BLOCKING AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Atracurium	1,000 mg/100 mL (10 mg/mL)	Load: 0.3 mg/kg Start: 5 mcg/kg/min Titrate: 2.5-5 mcg/kg/min. every 10 minutes	30 mcg/kg/min	Central or peripheral administration Restricted to ICU and AES Initial load may be repeated if clinically indicated
Cisatracurium (Nimbex)	200 mg/100 mL (2 mg/mL)	Load: 0.1 mg/kg Start: 3 mcg/kg/min Titrate: 1 mcg/kg/min. every 30 minutes	10 mcg/kg/min	Central or peripheral administration Restricted to ICU and AES Initial load may be repeated if clinically indicated
Vecuronium	50 mg/250 mL (200 mcg/mL)	Load: 0.1 mg/kg Start: 1 mcg/kg/min Titrate: 0.1 mcg/kg/min. every 10 minutes	2 mcg/kg/min	Central or peripheral administration Restricted to ICU and AES Initial load may be repeated if clinically indicated
VASOPRESSOR	STANDARD	STARTING DOSE/	MAXIMUM	NOTES

AGENTS	CONCENTRATION	TITRATION	DOSE	
Angiotensin II (Giapreza)	2.5 mg/250 mL (10,000 ng/mL)	Start: 20 ng/kg/min Titrate: 1.25 – 15 ng/kg/min. every 5 minutes	Initial 3 hours: 80 ng/kg/min Maintenance: Not to exceed 40 ng/kg/min	Central administration preferred Restricted to ICU and AES Attending approval required Patients must concurrently be receiving 0.25 mcg/kg/min. of norepinephrine equivalents
DOPamine	800 mg/250 mL (3.2 mg/mL)	Start: 2-5 mcg/kg/min Titrate: 1-5 mcg/kg/min. every 1 minute	20 mcg/kg/min	Central administration preferred for doses over 5 mcg/kg/min.
EPINEPHrine	5 mg/250 mL (20 mcg/mL)	Start: 0.01 mcg/kg/min Titrate: 0.01-0.2 mcg/kg/min. every 1 minute	1 mcg/kg/min CVICU: 0.2 mcg/kg/min	Central administration preferred
Norepinephrine (Levophed)	16 mg/250 mL (64 mcg/mL)	Start: 0.1 mcg/kg/min CVICU: 0.01 mcg/kg/min Titrate: 0.01-0.2 mcg/kg/min. every 1 minute CVICU: 0.01-0.2 mcg/kg/min. every 1 minute	1 mcg/kg/min CVICU: 0.2 mcg/kg/min	Central administration preferred
Phenylephrine (NeoSynephrine)	50 mg/250 mL (200 mcg/mL)	Start: 50-100 mcg/min CVICU: 5-100 mcg/min Titrate: 20-50 mcg/min. every 1 minute	300 mcg/min CVICU: 100 mcg/min	Central administration preferred
VAsopressin (Pitressin)	40 units/40 mL (1 unit/ml)	Start: 0.01-0.03 units/min. CVICU: 1 unit/hour Titrate: Per provider instructions CVICU: 1-5 units/hour	0.06 units/min. for sepsis or 0.08 units/min. for cardiac CVICU: 5 units/hour	Central administration preferred

HYPERTENSION & RATE CONTROLLING AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Clevidipine (Cleviprex)	25 mg/50 mL (0.5 mg/mL)	Start: 1-2 mg/hour Titrate: 1-2 mg/hour every 90 seconds by doubling the dose	16 mg/hour 1,000 ml/24 hours due to lipid base	Central or peripheral administration
Diltiazem (Cardizem)	100 mg/100 mL (1 mg/mL)	Load: 5-20 mg Start: 5 mg/hour Titrate: 2.5-5 mg/hour every 30 minutes	15 mg/hour	Central or peripheral administration
Esmolol (Brevibloc)	2,000 mg/100 mL (20 mg/mL)	Start: 25-50 mcg/kg/min Titrate: 25-50 mcg/kg/min. every 2 minutes	300 mcg/kg/min	Central or peripheral administration
Labetalol (Normodyne)	500 mg/250 mL (2 mg/mL)	Load: 10-20 mg Start: 0.5-1 mg/min. Titrate: 0.5 mg/min. every 30 minutes	3 mg/min	Central or peripheral administration
NiCARDipine (Cardene)	40 mg/200 mL (0.2 mg/mL)	Start: 2.5-5 mg/hour Titrate: 2.5 mg/hour every 15 minutes	15 mg/hour	Central administration preferred (if peripheral, rotate the infusion site every 12 hours)
HYPERTENSION & RATE CONTROLLING AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Nitroglycerin	100 mg/250 mL (0.4 mg/mL)	Start: 5-10 mcg/min Titrate: 5-10 mcg/min. every 2 minutes	200 mcg/min	Central or peripheral administration
Nitroprusside (Nipride)	50 mg/100 mL (0.5 mg/mL)	Start: 0.2-0.5 mcg/kg/min Titrate: 0.2-0.5 mcg/kg/min. every 2 minutes	10 mcg/kg/min	Central or peripheral administration
ANTIARRHYTHMIC AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Amiodarone (Cordarone)	600 mg/500 mL (1.2 mg/mL) <i>Mix in D5W only</i>	Load: 300 mg rapid IVP for pulseless VT	1 mg/min	Central administration preferred (if

		150 mg over 10 minutes for VT with a pulse or A-Fib (may repeat x1) Start: 1 mg/min. for 6 hours followed by 0.5 mg/min. for 18 hours		peripheral, avoid wrist or hand and infuse through a large vein) Requires the use of a 0.22 micron in line filter Maximum daily dose is 2 gm
Lidocaine	2,000 mg/250 mL (8 mg/mL)	Load: 1-1.5 mg/kg – may repeat in 5 minutes Start: 0.5 mg/min Titrate: Per provider instructions	4 mg/min	Central or peripheral administration Monitor serum lidocaine levels 12-24 hours after initiation (goal 1.5-6 mg/L)
Procainamide	2,000 mg/250 mL (8 mg/mL)	Load: 15-18 mg/kg over 25-30 min. or 100-200 mg repeated every 5 minutes as needed to a total dose of 1 gram Start: 1 mg/min Titrate: Per provider instructions	4 mg/min	Central or peripheral administration Reduce loading dose to 12 mg/kg in severe renal or cardiac impairment
DIURETIC AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Bumetanide (Bumex)	12.5 mg/50 mL (0.25 mg/mL)	Start: 1-4 mg/hour Titrate: 0.5-1 mg/hour every 2 hours	5 mg/hour	Central or peripheral administration
Furosemide (Lasix)	500 mg/50 mL (10 mg/mL)	Start: 5-10 mg/hour Titrate: 5-10 mg/hour every 2 hours	200 mg/hour	Central or peripheral administration
INOTROPIC AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
DOBUtamine	1,000 mg/250 mL (4 mg/mL)	Start: 1-2.5 mcg/kg/min CVICU: 1-2.5 mcg/kg/min Titrate: 2.5 mcg/kg/min. every 5 minutes CVICU: Per provider instructions	20 mcg/kg/min CVICU: 10 mcg/kg/min	Central administration preferred
Isoproterenol	2,000 mcg/250	Start: 0.5 mcg/min	10 mcg/min	Central

(Isuprel)	mL (8 mcg/mL)	Titrate: Per provider instructions	<i>CVICU: 4 mcg/min</i>	administration preferred
Milrinone (Primacor)	20 mg/100 mL (0.2 mg/mL)	Start: 0.125-0.25 mcg/kg/min Titrate: Per provider instructions	0.75 mcg/kg/min	Central administration preferred
Naloxone (Narcan)	2 mg/500 mL (0.004 mg/mL)	Start: 0.25-1 mg/hour Titrate: 0.25-1 mg/hour every hour	No maximum rate	Central or peripheral administration
INOTROPIC AGENTS	STANDARD CONCENTRATION	STARTING DOSE/ TITRATION	MAXIMUM DOSE	NOTES
Neostigmine	5 mg/50 mL (0.1 mg/mL)	Start: 0.4 mg/hour Titrate: Per provider instructions <i>No bolus dosing</i>	0.8 mg/hour	Please see: Neostigmine Guidelines

Summary of Changes

- PVSC: new vasopressin presentation (40 units/ 40 mL), not a concentration change
- Aligned loop diuretic max doses with Telemetry grid (approved MedSafe 7/2023)
- Fenoldopam, nesiritide, torsemide removed from US market, so removed from this document
- Relinked neostigmine guideline from PharmWeb to PolicyStat
- Added LASA tallMAN lettering
- Added ketamine
- Removed stray formatting
- Changed strength/concentration to USP7 convention

References

None

Exhibits/Attachments

None

Document Approval & Tracking

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Official Approval(s)	On PolicyStat Approval Workflow	

Approval Signatures

Step Description	Approver	Date
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PolicyStat Administrator Review	Olivia Curl: Admin Specialist Inter Health	8/2/2023
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